

Institute of Actuaries of India

Subject SP2 – Life Insurance Principles

November 2020 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- The lapse assumption should reflect the expected future experience in respect of the term assurance plan.
- They will be based on an analysis of company's past experience.
...how many years in the past should depend on the credibility of the past data
... going too much into the past may not be relevant. The right balance has to be taken for experience analysis.
- If no such experience exists or the company data is inadequate then the experience under any similar contracts would be analysed.
- If the company does not have adequate data, there may be industry wise experience it could use.
- Lapses can be analysed by various parameters:
 - The most important being duration, since lapse rates tend to vary by duration at least for initial few years.
 - By sales channel (direct, thro intermediaries, etc)
 - Other factors could be
 - age and gender
 - premium size
 - premium frequency or mode of payment
 - risk categories (smoker/ non-smoker)
- the detail of the analysis depends on the volume of credible data available under each of this subsets
- The persistency rate can be defined as number of policies surviving the year divided by number of policies inforce at the start of the year.
- And lapse rate for each year could be 1 minus persistency rate of that year.
- Persistency could also be calculated as a cumulative figure as number of policies surviving the year dividing by the number of policies issued at outset.
- Also compare these with the latest lapse assumptions taken in pricing.
- The calculated persistency rates may be adjusted for :
 - Any economic recession
 - Difference in the target market, sales channel (online vs offline)
 - Any change in the processes (sales process, underwriting process)
 - Any change in benefit structure or level (surrender values payable if very different from the current practice)
 - How competitor's product fares against ours
 - Any change in tax rates, new regulations, etc
- It is necessary to explore sensitivity of company's profits to variation in the future lapse experience.

[8 Marks]**Solution 2:** Parameters to monitor for maintaining a profitable annuity portfolio

- **Investment and investment returns**
 - (i) Asset liability matching exercise is carried out continuously; any change in asset liability matching due to change in experience compared to that assumed in pricing; will need readjustment.
 - (ii) The quality of assets in terms of default risk, market risk and liquidity risk are monitored continuously.
 - (iii) In case assets tenure was shorter than the liability tenure (due to the non-availability of very long term assets in the market); there is a need to watch-out for the reinvestment risk (as the annuity rates can't be reviewed); if needed get into forward yield arrangements
 - (iv) The yields on assets are monitored continuously.
 - (v) The overall investment and economic market of the country is monitored continuously.

- **Mortality**
 - (i) Lower level of mortality (or higher longevity) than anticipated may lead to lower level of profitability than anticipated.
 - (ii) Lower level of mortality will also increase service level expenses as annuity payments will happen for a longer duration than anticipated.
 - (iii) Lower mortality also means liabilities will run longer than the assets invested for; if the portfolio was originally matched. There is a reinvestment risk in that case.
- **Expenses**
 - (i) The level of expenses not to be more than anticipated leading to lower level of profitability.
 - (ii) Since these contracts are very long term, the level of expense inflation is also a critical factor in the profitability of the portfolio
 - (iii) One off expenses may be identified and impact of them on profitability is monitored.
 - (iv) Any change in tax rates.
- **Level of new business**
 - (i) This is monitored to understand whether the business written is in line with the anticipated at the time of profit testing.
 - (ii) Lower than anticipated business may lead to increase in per policy expenses. On the other hand, higher than anticipated business may lead to higher capital requirements. Both may be deterrent to the profitability of the contract.
 - (iii) The assumption with respect to mortality also works better if there is a sizeable volume of business and not a diminishing portfolio
- **Reserving and capital requirement:**
 - (i) Regulator may tighten the reserving requirement and solvency margin requirement leading to impact on profitability from time value perspective.

Monitoring of new business strain is very crucial if capital is a constraint.

[9 Marks]

Solution 3:

- i) For unit linked contracts, lower level of prudence in the reserving basis would decrease the requirement for non-unit reserves

...thereby increasing the net assets compared to prudent basis.

- As non-unit reserves are lower, the shareholders value resulting from release of non-unit reserves over the projected future period will be smaller than the release under prudent basis.
- However, this impact is smaller compared to the impact on net assets.
- Further, if the solvency capital is computed based on reserves then the overall solvency capital will reduce and hence cost of solvency capital will also reduce.
- Thus, the net impact is increasing the embedded value when less prudence is used in the reserving basis.

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ii) Embedded Value

Given

In Crores		In Crores	
Assets	10000	Unit Reserves	6000
		Non-Unit Reserves	500
		Solvency Capital	120

Net assets (Adjusted Net worth)
(10000-(6000+500)) **3,500**
Crores

i.e. (Free Surplus+ solvency Capital)

Projected Charges and expenses (in Crores)

	Year end					
Time	Unit reserves	Charges	Expenses	Net cashflow to Shareholders		P.V.
1	6,480.00	64.80	32.40	32.40		28.93
2	4,618.94	46.19	23.09	23.09		18.41
3	2,469.29	24.69	12.35	12.35		8.79
					Total	56.13

Explanation:

Year 1: Unit reserves (year end) = 6000 * 1.08 = 6480

Charges = 6480 * 1% = 64.80

Expenses = 6480 * 0.5% = 32.40

Net Cashflow = Charges – Expenses = 64.80-32.40=32.40

Year 2: Unit reserves (year end) = (0.6666/1)*(6480-64.80) * 1.08 = 4618.94

Charges = 4618.94 * 1% = 46.19

Expenses = 4618.94 * 0.5% = 23.09

Net Cashflow = Charges – Expenses = 46.19-23.09 = 23.09

Year 3: Unit reserves (year end) =(0.3333/0.6666) (4618.94-46.19) * 1.08 = 2469.29

Charges = 2469.29 * 1% = 24.69

Expenses = 2469.29 * 0.5% = 12.35

Net Cashflow = Charges – Expenses = 24.69-12.35=12.35

$$\text{PV of net cashflow from unit fund} = \frac{32.40}{1.12} + \frac{23.09}{1.12^2} + \frac{12.35}{1.12^3} = \mathbf{56.13 \text{ Crores}}$$

Release of Non-Unit reserves (in Crores)

Time	Non-Unit reserves (Start)	Interest on Non-Unit reserves	Non-Unit Reserve release due to maturity	Total cashflow		PV of net cashflow from non-unit fund
1	500	40.00	166.67	206.67		184.52
2	333.33	26.67	166.67	193.33		154.12
3	166.67	13.33	166.67	180.00		128.12
					Total	466.77

Interest on non unit reserves = Non unit reserves at the start *1.08

As 1/3rd of policies are getting matured in the each of the next three years and hence non-unit reserves of 500/3 = 167 is released in every year for the next three years.

Total Cashflow = Interest on non-unit reserves + Non unit reserve release due to maturity

$$\text{PV of net cashflow from non-unit fund} = \frac{206.67}{1.12} + \frac{193.33}{1.12^2} + \frac{180.00}{1.12^3} = \mathbf{466.77 \text{ Crores}}$$

Embedded Value = Net asset + PV of charges from unit fund plus release of non-unit fund.

$$\text{Embedded Value} = 3500 + 56.13 + 466.77 = 4023 \text{ Crores}$$

[10]

[13 Marks]

Solution 4:

Due to the market volatility, assuming that the insurance company is also subjected to the loss of 30% to 40% of market value:

This fall could be a short term dip and there could be no fundamental deterioration in the market

... in such case the market may be expected to correct itself after sometime.

The possible major impact could be to valuation of assets on market value for the financial statement at the year end.

If market volatility is due to something fundamentally wrong that has come to light which reflected through the market dip:

.... this effect may be expected to continue for a longer period.

.... the reduction in the real value of these stocks may have a significant impact on the insurer's portfolio.

Possible impact of market volatility on the insurance companies**i) Unit Linked Insurance Products:**

- a) On unit-linked business the investment risk is mainly borne by the policyholder.
- b) Except in case of any benefit guaranteed, either financial guarantee or non-unit related guarantee, the insurer will have to meet the excess of any guaranteed benefit over the value of the units.
- c) The nature and extent of any financial risks from investment, expenses and demographic assumptions will depend on the nature and level of any guarantees given.
- d) charges linked to fund size like fund management charge will be reduced by poor investment performance resulting into insufficient income to meet its expenses and may impact profitability.
- e) The ability to review the charges and any constraints imposed on the level of charges by marketing or legislative requirements will exacerbate the risk.
- f) The insurance company may be subjected to reputational risks, due to reduced fund values and may impact future sales.
- g) If this is true across industry, the effect may affect the whole industry.
- h) if a particular insurer has taken a higher hit compared to others, then that company will suffer higher reputational and future new business damage
- i) The insurance company may also be subjected to surrender risk; because of loss of confidence in insurance products Or due to panic among policyholders.

[4]

ii) With profits insurance products:

- a) The insurance company may have some investment risk because of the guaranteed sum assured and any reversionary bonuses already declared over the term.
- b) The investment risk is rather lower on with-profits contracts. If investment returns are lower than expected then the insurance company may be able to reduce bonuses, passing some of the investment risk back to the policyholder.
- c) On the other hand, this risk can be absorbed through the estate and without affecting declaration of regular reversionary bonuses

.... declaration of terminal bonuses for policies becoming claims could be reduced.

- d) Also since it is long term, there is a possibility of the insurer taking corrective action to reduce the impact of the market fall
- e) However, there could be still a guaranteed element that the insurance company must meet.
- f) Policyholder expectations may limit the scope and speed of bonus reductions.
- g) It is possible that due to reduced bonuses, corresponding shareholders share of surplus distribution may also be reduced.
- h) Shareholders share of surplus may also be reduced in absolute level, in line with the declaration of reduced level of bonuses.
- i) If bonuses declared are much lower than the PRE while making benefit pay-outs, the insurer could be subjected to regulatory risk.

[4]

iii) Index linked insurance products:

- a) Under the index linked contracts the benefits are directly related to some underlying index.
- b) The impact on this portfolio depends on the correlation of the benchmarked index to the equity market movement
- c) For example if the index to be matched was a bond index, like "10year G-Sec GRY" then there is a chance that due to the equity market fall, the bonds might have improved on yield. It might be actually good for the customers or otherwise.
- d) However, if the benchmark index was an equity index; then the impact could be direct
- e) Again the extent of the impact on the policies would depend on how the benefits are designed. Say if the benefits are dependent on the 3 months or 6 months average of the index; the last 10 days fall would get a bit absorbed in the average.
- f) The impact can be different if there is any form of mismatch between the actual assets held and the index composition
- g) If the stocks held by the insurer have performed poorly compared to the index, the insurer may be subjected to higher investment risk.
- h) Similar to unit linked insurance products; insurance company may also be subjected to the loss of income, if charges are linked to the underlying fund resulting in insufficient charges to meet the expenses and profits. The effect may be a little lower compared to ULIP, as the benefit declaration would be smoothed and not marked-to market like in case of unit linked, if the product design has averaged it.
- i) Reputational risk, competition risk and loss of sales relatively higher, when compared to other products, due to the underlying investment guarantee in the product.

[4]

iv) Without profit insurance products:

- a) The main risk to the insurance company relates to investment.
- b) Since under without profit contracts all the benefits are guaranteed, the proportion of equity investment would be lower.
- c) Without profit contracts like term assurance or any pure risk portfolios will have a very small reserve to invest; so the impact could be very low
- d) Non-par savings products where all the benefits are guaranteed; the chances are that the guarantee in the liability was met with the fixed income assets and any equity investment was meant to give shareholders their profit for taking higher risk.
- e) In such a case the shareholders profit will take a hit
- f) Again the final impact of hit depends on if the company needs to sell the assets in the near future.
- g) If the equities can be held for long term and dispose them at some opportune time; then the impact to the shareholders can also be kept low

[3]

[15 Marks]

Solution 5:

- i) Cost of capital should be allowed for in the arriving at the asset share calculations, as:
Asset share is the accumulation of premiums less deductions associated with the contract, all accumulated at the actual rate of interest earned on investments.

The deductions associated with the contract also include cost of any capital necessary to support the contract.

- a) The insurance company makes a loss on writing a new policy due to the new business strain.
- b) In order to write the new business, insurance companies need capital to fund the new business strain.
- c) The capital is thus, comes either from the surplus built up over previous years from other policies but not yet distributed, and/or from the shareholders' free assets.
- d) These other policyholders, and shareholders, have in effect "invested in" the new policy by putting up the money to fund the new business strain. They therefore require some money back from this investment.
- e) In order to meet the cost of financing the new business strain, a deduction from the asset share of a policy is made to pay back to the other policyholders/shareholders.
- f) The shareholders will get money back from with-profits policyholders through the shareholders' transfers. This reflects the necessary return on their capital.
- g) The capital provided by policyholders cannot be easily assigned to individual existing policies, particularly as most of the capital has come from policies that are no longer in force.
- h) To reflect the cost of this financing by other policyholders, insurer may reduce or increase its existing capital supply, and so may make further adjustments to existing policies' asset shares to this effect.
- i) During early duration, deductions may be made from the asset shares to pay for the capital being required at those times. However, further additions or deductions at later durations, reflect the company's overall policy to expand or contract the company's capital base in addition to any retention of shareholders' profits for this purpose.
- j) The main reason behind the adjustments made to asset shares is to ensure that the different groups of with-profits policyholders are treated fairly.
- k) Thus the surplus which eventually accrues in respect of any with-profits policy should be reduced by some "cost of capital" to represent that policy paying something in respect of new business strain.

[6]

ii)

- a) Margins for future adverse experience:

- While pricing a with participation product, insurance company may include explicit or implicit margins both for covering any adverse deviations and to generate profit.
- The existence of these implicit or explicit margins entails profit distribution to policyholders
- In a with profit contract,
 - the base benefit is only guaranteed and
 - the release of the margins are used to distribute bonus/dividends to policyholders.
- That way the policyholder also participates in the profits of the portfolio experience.

- On the flip side if the experience worsens the bonus/ dividend could be reduced from the past declarations.
- However, to what extent can the company reduce bonuses using these margins depends on the reasonable expectations of policyholders in this respect.
- Hence, it very difficult for insurance companies to reflect poor experience fully in the bonuses they declare.
- Also guarantees under all with-profits policies mean that there is some level of adverse experience beyond which any further losses cannot be recouped.
- There would also be guidelines for smoothing the bonus declarations; internal to the company or by regulations.
- There would also be competitive pressure on what other companies are doing and marketing risk of attracting more new business, if the company reduced its bonus rates drastically.

[5]

b)

Treatment under each method of bonus distribution:

- Under revalorisation method, insurer:
 - Declares only investment profits, in general, and any expense or mortality losses will have to be borne by the company.
 - must declare bonuses in accordance with the surplus accrued. Hence, will have to reflect the experience.
- For Contribution method and additions to benefit method:
 - generally, profits from all sources are distributed amongst both policyholders and shareholders.
 - Hence, any loss in any of the source could impact both the policyholders and shareholders.
 - In any year, if the surplus is very low or very high and this was seen not as a regular event, the bonus rates might be kept unchanged i.e., declare more or less than indicated by asset share projections depending on the situation

[4]

[15 Marks]

Solution 6:**i) Difference between the unit linked products and unitised with profit products:**

Feature	Unit linked	Unitised with profit
Unit pricing	Based on the value of the underlying asset. Is calculated daily. There is no discretion to the insurer in the unit price calculations The benefit paid to the policyholder is solely the bid value of the units held less any penalty or charge on exit.	There is no direct link to the pool of assets There could be two ways: The price of the unit remains constant & the company allocates additional units to each contract, as part of the bonus declaration. Instead they may decide to change the unit price daily towards the bonus declared, without adding more units to the policy The key difference is there is discretion in the declaration Company takes long term view of the investments return A terminal bonus may also be added at the time of a claim. So the benefit paid is not solely based on the bid value of the units Bonuses added is smoothed
Benefit payable on surrender	It will be the bid price of the units less any surrender penalty	Here also it is the bid price of the units less any surrender penalty less any market value adjustment (MVA).
	There is no discretion with the insurer	The size of the MVA is at the discretion of the insurer
Charges	There are explicit charges like Policy allocation charge Premium allocation charge Fund management charge Mortality charge Bid-offer spread	Unitised with profit could have explicit charges like in case of unit linked contract or have implicit charges in the bonus declarations

[9]

ii) Unintended change in the mix of new business by nature or size of the contract:

- a) Generally, any change in the mix by nature for this insurance company might involve, a change in the mix by:

- class of business (proportion of a particular class of business like unit linked),
 - type of contract (for e.g. in this case - Unit linked versus unitised with profits),
 - contract design (for e.g. in this case unit linked versus with profits),
 - premium frequency (for e.g single premium versus regular premium).
- b) The risk profile of the customers buying with profit, unit linked and unitised with profit contracts are different, based on the risk appetite of the customer. Hence the risk profile of each of these portfolios varies for the insurer.
- c) Smaller premium size policies may be lower mortality risk however they may not be contributing to the per policy expenses so much.
- d) The new business strain for each of this product type will vary based on the charging structure or the benefit design of the product. Any change in the mix of business from that anticipated will cause a change in the capital requirement
- e) In particular, the risk profile and the capital required under the with-profit products depend on the initial guarantees offered and the bonus distribution philosophy of the insurance company. This may be similar to the unitised with profit products also.
- f) With regard to the unit linked products, the risk profile and the capital requirements depend on the charging structure, guarantees, if any offered, the ability to increase the charges and how much of the charges can be matched to the higher initial commission and expenses.
- g) If there is a significant unintended change in the mix of the business actually written, it could lead to mismatch between the charges accruing to the company and its actual expenses.
- h) A mismatch between charges and expenses might result from a change in the mix of contracts by nature and size.
- i) For example, under the unit linked policies, if insurance company could not achieve its business targets or if all the policies sold are of smaller size premiums, the charges it may accrue may be relatively low and its contribution to total expenses from a particular type of contract could be lower.
- j) If the mix of business continues, the insurance company might not receive a sufficient overall contribution to cover its expenses. This may result in capital requirement.
- k) The regulatory solvency requirement for each of the product types could be different based on the guarantees offered under this plans. This also can be a cause for change in the capital requirement with change in business mix.
- l) Hence, any significant unintended change in the mix of new business by nature or by size of contract could lead to a significant change in the risk profile or capital needs of the company that were not within the resources available to it.

[9]

[18 Marks]**Solution 7:**

- i) Due to the pandemic, the insurer is expected to have higher claims experience this year compared to earlier years on his portfolio.
- In case pandemic is an exclusion in the product:
 - (i) insurer is not bound to accept claims due to Pandemic
 - (ii) So there may be no need to revisit its existing reinsurance arrangements; however if the competitors are covering; it may need to revisit the coverage for pandemic
 - (iii) If future expected mortality is expected to be impacted overall there may be a need to re-price and hence review the appropriateness of the existing reinsurance arrangements.
 - (iv) The infection could exaggerate the mortality rate due to other reasons (like kidney failure, lung infection, etc) without actually being the reason for death. So the exclusion won't be possible to apply.

- Pandemic cover included in the product:
 - (i) If included in the policy contract, but excluded in the corresponding reinsurance arrangement:
 - Need to revisit the existing reinsurance arrangements
 - (ii) If both policy contract and corresponding reinsurance arrangement includes the pandemic cover:
 - Still revisit to ensure that the reinsurance arrangements are appropriate to the current context.
- Revisiting the reinsurance arrangements:
 - (i) Due to the pandemic, the insurer may expect adverse claims experience
 - ...Increased claims experience may also warrant higher reserving and possibly additional capital
 - ...expect some profit volatility, depending on the size of the risk portfolio
- The possible reinsurance support the insurer could be looking for its pure protection portfolio could be to:
 - (i) Depending on the distribution of the existing policyholders (by age, SA, social strata), there is a risk of many small claims
 - ... in which case the standard surplus reinsurance will not be effective
 - (ii) An excess of loss reinsurance contract could provide better coverage to limit the exposure on the total claims occurring within the portfolio.
 - ... to ensure that the excess of loss covers don't exclude pandemic.
 - (iii) If the insurer expects the impact of the pandemic to hit its financial soundness, it may opt for any capital financing
 - either in the form of quota share arrangement or
 - a contingent loan or
 - contingent on future supervisory profits from the business.
- Administration support
 - (i) If there are constraints on testing the disease (lesser testing kits, delay in results, limited centers to test, etc), there could be some death which are not classified as due to the pandemic. Reinsurer should support accepting such cases
 - (ii) There could be more stress on claims processing, so need assistance from the reinsurer for faster claims approval either by adding more man-power
 - (iii) Or support through online claims approval system.
 - (iv) Accepting large risks - Capacity to accept large risks may be enhanced (up to a maximum) to be accepted automatically to ease smooth claims underwriting process.
- Technical Support
 - For existing customers

The reinsurer will be able to share best practices followed in other countries (as this pandemic has effect other countries too) in terms of

 - .. customer education with respect to prevention
 - ... care taking incase of infection
 - For New business

If it lacks experience data and expertise in handling pandemic, assistance for underwriting, pricing and product design may also be obtained from reinsurers.

ii)

- a) The insurer should offer its existing policyholders the option to add pandemic cover under their existing policy itself.

Possible challenges the insurer may face in designing:

- (i) Should it be offered under all plans or only savings plans or term plans
- (ii) Offering them under savings contract may be less risky
- (iii) there may be a risk of how many will opt for the option
- (iv) Less takers - possible anti –selection
- (v) Could be because of inappropriate marketing strategies.
- (vi) Time available to avail the option –less time lesser anti-selection and vice versa
- (vii) Will this be one time option -less risky & easy to administer
- (viii) or flexibility given to the policyholder to exercise the option at various events – could prove to be risky and may need continuous support to administer the option.
- (ix) Time and effort the insurer needs to put in educating the existing customers to avail this option
- (x) Underwriting considerations – should the option be open to all or for only standard lives – sub-standard lives may pose higher mortality risk.
- (xi) Reinsurer support –existing reinsurance arrangements may not be suitable and may need additional protection, policy data on pandemics and possible technical support on underwriting.
- (xii) The Mode of reaching out the offering of the option to the current customers – corresponding influence on the mortality experience and the associated costs

[5]

b)

Possible challenges in pricing the option:

- Policy and other data :
 - (i) Availability of appropriate and relevant data could be challenging.
- Mortality risks:
 - (i) May have to depend on population data for general incidence of infection and for mortality rate;
 - (ii) May also take support from reinsurer;
 - (iii) Arriving at future expected experience could be difficult, as it has not yet completely evolved;
 - (iv) Leading to insufficient data, model and parameter risk;
 - (v) If the option take up rate is lower – increases random fluctuations risk;
 - (vi) The extent of these risks will depend on the reliability and applicability of data available;
 - (vii) Risk of inappropriate underwriting, given the possible limitation to access medical facilities due to pandemic;
- Risks from investment:
 - (i) As the size of the reserves expected to build up and the underlying investment guarantees are small, the risk may generally be low;
 - (ii) However, market volatility, if any due to pandemics, may generally impact the overall investment portfolio;
 - (iii) May lead to general investment risk, though the impact could be very minimal.
- Regulation risks:
 - (i) Insurers may face regulatory risk if the pricing is viewed as unreasonably high.

- Expenses risk:
 - (i) Depending on the flexibility in the timing to avail the option, the level of underwriting and the associated costs, the expense risk may vary;
 - (ii) the cost to medically test for the infection could increase the expenses assumed
 - (iii) To reach out to the customers during the pandemic, may require additional efforts & cost;
 - (iv) If option take up rate is low, the option cost per policy may appear higher
- Non-availability of data - feasibility of the new option and possible insufficient profitability – given that already pandemic has hit.
- Option taking up rate could be significantly affected by what the competition is doing.
- Keeping up with the competition can lead to increased risk in other areas, for example:
 - (i) Reduction in profit margins in the pricing basis
 - (ii) Insufficient loadings for expenses
 - (iii) Relaxed underwriting requirements

[8]

[22 Marks]
